

Jonah Rose

Curriculum Vitae

Department of Physics
Princeton University
☎ (+1) 9714009332
✉ jr8952@princeton.edu
🌐 My Webpage
ORCID: 0000-0002-2628-0237

Education

- 2024–present **Postdoc**, *Princeton*.
Developing The DREAMS Project
- Fall 2023 **Predoctoral Fellowship**, *Center for Computational Astrophysics*.
Peering Into the Black Box - informed astrophysical analyses with machine learning
- 2019–2024 **Doctor of Philosophy: Astronomy**, *University of Florida*.
Cosmological simulations, alternative dark matter, machine learning
- 2019–2021 **Master of Science: Astronomy**, *University of Florida*.
GPA: 3.97
Thesis: *Where Binary Neutron Stars Merge: Predictions from IllustrisTNG*
- 2015–2019 **Bachelor of Science: Physics**, *University of Oregon*.
Upper-Division GPA: Physics - 3.82
- Bachelor of Science: Computer Science & Math**, *University of Oregon*.
Upper-Division GPA: Math - 3.93; CS - 4.00

Refereed Publications

- 2026 Ilem Leisher, Paul Torrey, Alex M. Garcia, **Rose, Jonah C.**, Francisco Villaescusa-Navarro, Zachary Lubberts, Arya Farahi, Stephanie O'Neil, Xuejian Shen, Olivia Mostow, Nitya Kallivayalil, Dhruv Zimmerman, Desika Narayanan, and Mark Vogelsberger. Linking Warm Dark Matter to Merger Tree Histories via Deep Learning Networks. , volume 1005, page 186, July 2026.
- 2026 Ethan Lilie, **Rose, Jonah C.**, Mariangela Lisanti, Alex M. Garcia, Paul Torrey, Kassidy E. Kollmann, Jiaxuan Li, Olivia Mostow, Bonny Y. Wang, Stephanie O'Neil, Xuejian Shen, Alyson M. Brooks, Arya Farahi, Nitya Kallivayalil, Lina Necib, Andrew B. Pace, and Mark Vogelsberger. The DREAMS Project: Disentangling the Impact of Halo-to-Halo Variance and Baryonic Feedback on Milky Way Dark Matter Speed Distributions. , volume 1002, page 168, May 2026.
- 2026 Alex M. Garcia, Paul **Rose, Jonah C.**, Andrea Caputo, Mariangela Lisanti, Andrew B. Pace, Hongwan Liu, Abdelaziz Hussein, Haozhe Liu, Francisco Villaescusa-Navarro, John Barry, Ilem Leisher, Belén Costanza, Jonathan Kho, Ethan Lilie, Jiaxuan Li, Niusha Ahvazi, Aklant Bhowmick, Tri Nguyen, Stephanie O'Neil, Xiaowei Ou, Xuejian Shen, Arya Farahi, Nitya Kallivayalil, Lina Necib, and Mark Vogelsberger. The DREAMS Project: Disentangling the Impact of Halo-to-halo Variance and Baryonic Feedback on Milky Way Dark Matter Density Profiles. , volume 1002, page 8, May 2026.
- 2026 Ryan Low, Rakshak Adhikari, **Rose, Jonah C.**, Stephanie O'Neil, Mikhail V. Medvedev, Paul Torrey, and Mark Vogelsberger. Structure formation under inelastic two-component dark matter: halo statistics and matter power spectra in the high- z universe. , volume 546, page 2259, February 2026.
- 2026 Tri Nguyen, Francisco Villaescusa-Navarro, Siddharth Mishra-Sharma, Carolina Cuesta-Lazaro, Paul Torrey, Arya Farahi, Alex M. Garcia, **Rose, Jonah C.**, Stephanie O'Neil, Mark Vogelsberger, Xuejian Shen, Cian Roche, Daniel Anglés-Alcázar, Nitya Kallivayalil, Julian B. Muñoz, Francis-Yan Cyr-Racine, Sandip Roy, Lina Necib, and Kassidy E. Kollmann. How DREAMS Are Made: Emulating Satellite Galaxy and Subhalo Populations with Diffusion Models and Point Clouds. , volume 997, page 336, February 2026.

- 2026 M. Silvestrini, C. Tortora, V. Busillo, Alyson M. Brooks, A. Farahi, A. M. Garcia, N. Kallivayalil, N. R. Napolitano, **Rose, Jonah C.**, P. Torrey, F. Villaescusa-Navarro, and M. Vogelsberger. CASCO: Cosmological and ASTrophysical parameters from Cosmological simulations and Observations: IV. Testing warm dark matter cosmologies with galaxy scaling relations: A joint simulationobservation study using DREAMS simulations. , volume 706, page A382, February 2026.
- 2025 Olivia Mostow, Paul Torrey, **Rose, Jonah C.**, Alex M. Garcia, Niussha Ahvazi, Mariangela Lisanti, and Nitya Kallivayalil. How Many Bursts Does It Take to Form a Core at the Center of a Galaxy? , volume 995, page 25, December 2025.
- 2025 Belén Costanza, Bonny Y. Wang, Francisco Villaescusa-Navarro, Alex M. Garcia, **Rose, Jonah C.**, Mark Vogelsberger, Paul Torrey, Arya Farahi, Xuejian Shen, and Ilem Leisher. On the Sensitivity of Different Galaxy Properties to Warm Dark Matter. , volume 994, page 62, November 2025.
- 2025 **Rose, Jonah C.**, Paul Torrey, Francisco Villaescusa-Navarro, Mariangela Lisanti, and the DREAMS Collaboration. Introducing the DREAMS Project: DaRk mattEr and Astrophysics with Machine learning and Simulations. *ApJ*, volume 982, page 68, April 2025.
- 2024 Shurui Lin, Francisco Villaescusa-Navarro, and **Rose, Jonah** et. al. Can We Constrain Warm Dark Matter Masses with Individual Galaxies? *APJ*, volume 970, page 170, August 2024.
- 2024 Hannah Richstein, Nitya Kallivayalil, Joshua D. Simon, Christopher T. Garling, Andrew Wetzel, Jack T. Warfield, Roeland P. van der Marel, Myoungwon Jeon, and **Rose, Jonah C.** et. al. Deep Hubble Space Telescope Photometry of Large Magellanic Cloud and Milky Way Ultrafaint Dwarfs: A Careful Look into the Magnitude–Size Relation. *Apj*, volume 967, page 72, May 2024.
- 2024 **Jonah C. Rose**, Paul Torrey, Francisco Villaescusa-Navarro, Mark Vogelsberger, Stephanie O’Neil, Mikhail V. Medvedev, Ryan Low, Rakshak Adhikari, and Daniel Anglés-Alcázar. Inferring Warm Dark Matter Masses with Deep Learning. *MNRAS*, volume 527, pages 739–755, January 2024.
- 2023 Stephanie O’Neil, Mark Vogelsberger, Saniya Heeba, and **Jonah C. Rose**, et. al. Schutz Katelin. Endothermic self-interacting dark matter in Milky Way-like dark matter haloes. *MNRAS*, volume 524, pages 288–306, September 2023.
- 2022 **Jonah C. Rose**, Paul Torrey, Mark Vogelsberger, and Stephanie O’Neil. Unravelling the interplay between SIDM and baryons in MW haloes: defining where baryons dictate heat transfer. *MNRAS*, volume 519, pages 5623–5636, December 2022.
- 2021 **Jonah C. Rose**, Paul Torrey, K. H. Lee, and I. Bartos. Where Binary Neutron Stars Merge: Predictions from IllustrisTNG. *APJ*, volume 909, page 207, March 2021.
- 2020 K. H. Lee, I. Bartos, G. C. Privon, **Jonah C. Rose**, and P. Torrey. FIRST J1419+3940 as the First Observed Radio Flare from a Neutron Star Merger. *APJ Letters*, volume 902, page L23, October 2020.

Pre-Print Publications

- 2025 **Jonah C. Rose** et al. ETHOS - an effective theory of structure formation: Small-Scale Tensions. *in prep.*, October 2025.
- 2025 Andreas Filipp, Tri Nguyen, Laurence Perreault-Levasseur, **Rose, Jonah C.**, Chris Lovell, Nicolas Payot, Francisco Villaescusa-Navarro, and Yashar Hezaveh. Predicting the Subhalo Mass Functions in Simulations from Galaxy Images. *arXiv e-prints*, page arXiv:2510.14766, October 2025.

- 2025 **Rose, Jonah C.**, Mariangela Lisanti, Paul Torrey, Francisco Villaescusa-Navarro, Alex M. Garcia, Arya Farahi, Carrie Filion, Alyson M. Brooks, Nitya Kallivayalil, Kassidy E. Kollmann, Ethan Lilie, Bonny Y. Wang, Akaxia Cruz, Sandip Roy, Andrew B. Pace, Niusha Ahvazi, Stephanie O'Neil, Cian Roche, Xuejian Shen, and Mark Vogelsberger. The DREAMS Project: A New Suite of 1,024 Simulations to Contextualize the Milky Way and Assess Physics Uncertainties. *arXiv e-prints*, page arXiv:2512.00148, November 2025.
- 2025 **Rose, Jonah C.**, Mariangela Lisanti, Paul Torrey, Francisco Villaescusa-Navarro, Alex M. Garcia, Arya Farahi, Carrie Filion, Alyson M. Brooks, Nitya Kallivayalil, Kassidy E. Kollmann, Ethan Lilie, Jiaxuan Li, Olivia Mostow, Akaxia Cruz, Tri Nguyen, Sandip Roy, Andrew B. Pace, Niusha Ahvazi, Stephanie O'Neil, Xuejian Shen, Francis-Yan Cyr-Racine, Adrian M. Price-Whelan, Marla Geha, Lina Necib, Mark Vogelsberger, Julian B. Muñoz, and Julianne J. Dalcanton. The DREAMS Project: Disentangling the Impact of Halo-to-Halo Variance and Baryonic Feedback on Milky Way Satellite Galaxies. *arXiv e-prints*, page arXiv:2512.02095, December 2025.

Research Experience

University of Florida

- Jun 2022 – present **Machine Learning with Alternative Dark Matter Models.**
Exploring how machine learning can be used to place constraints on alternative dark matter models, currently focusing on WDM.
- Advisor **Dr. Paul Torrey**, Assistant Professor, Department of Astronomy, University of Virginia
- Jul 2021 – Sep 2025 **Exploring the Validity of the ETHOS Model in Hydrodynamic Simulations.**
Running SIDM and ETHOS simulations with IllustrisTNG physics to understand if these models are viable to explain the small-scale problems.
- Advisor **Dr. Paul Torrey**, Associate Professor, Department of Astronomy, University of Florida
- Sep 2019 – Nov 2020 **Learning Where Binary Neutron Stars Merge.**
Using IllustrisTNG simulations to understand where binary neutron star mergers are most likely to occur to inform observational follow-ups of gravitational wave detections.
- Advisor **Dr. Paul Torrey**, Associate Professor, Department of Astronomy, University of Florida

University of Oregon

- Jan 2018 – Jun 2019 **The Assembly of Rich Clusters: A Wide-Field View of a Galaxy Protocluster at $z = 2.16$.**
Comparing galaxy cluster properties to nearby field galaxies.
- Advisor **Dr. Ricardo Demarco**, Associate Professor, Department of Astronomy, University of Concepción

Leadership, Mentorship, and Academic Service

Scientific Leadership & Workshop Organization

- Jul 2026 **DREAMS Work Day.**
Co-organized targeted pre-conference collaboration work day to align ongoing analyses and data generation projects.
- Sep 2025 **Working Group Lead.**
Restructured the DREAMS collaboration into specialized working groups to streamline research output.
- May 2025 **DREAMS Collaboration Workshop 2.**
Co-organized formal collaboration workshop.
- Dec 2023 **DREAMS Collaboration Workshop 1.**
Co-organized formal collaboration workshop.

Mentorship

- 2024 – Present **Graduate Student Mentor.**
Providing research guidance and ongoing informal mentorship to two doctoral candidates.

2022 **REU Student Mentor.**

Provided structured research guidance to an undergraduate REU student, culminating in a co-authored publication.

[Academic Service](#)

2023 - **Peer Reviewer.**

Present Referee for *Astronomy and Astrophysics* (A&A)

2023 - **Simulation Data Management.**

Present Oversee and manage large-scale simulation data access. Maintain guest collections to facilitate secure, high-performance data sharing across the collaboration and publicly.

2022-2023 **GAO Student Liaison.**

Gathered feedback from graduate students regarding environment, curriculum, and research to present to faculty.

[Fellowships & Awards](#)

Fall 2023 ***Center for Computational Astrophysics Predoctoral Fellowship***

2019 - 2024 ***University of Florida Graduate Student Fellowship***

2015 - 2019 ***Oregon Summit Merit Scholarship***

[Presentations](#)

Jul 2026 **Cosmic Horizons Conference** - University of Virginia

June 2026 **Simulation Based Inference for Galaxy Evolution 3** - Cambridge University (*Virtual*)

Jun 2026 **Advancing Field-level and SBI for Cosmology** - Perimeter Institute

Apr 2026 **CTC Seminar** - University of Maryland

June 2025 **SBI For Galaxy Evolution** - University of Bristol

May 2025 **DREAMS Workshop** - Flatiron Institute, Simons Foundation

Dec 2024 **CAMELS Workshop** - Flatiron Institute, Simons Foundation

Oct 2024 **Probes of Alternative Dark Matter Models** - University of Toronto

May 2024 **GECO Lecture Series** - UVA

Dec 2023 **Artificial Intelligence Tools to Study the Nature of Dark Matter** - Flatiron Institute, Simons Foundation

Nov 2023 **Debating the Potential of Machine Learning in Astronomical Surveys #2** - Flatiron Institute, Simons Foundation

Sep 2023 **Particle Astrophysics & Cosmology Meeting Around NYC** - Flatiron Institute, Simons Foundation

Jun 2023 **Self-Interacting Dark Matter: Models, Simulations and Signals** - Pollica Physics Center, Italy

Nov 2022 **CAMELS Workshop** - Flatiron Institute, Simons Foundation

[Computer Skills](#)

Programming Languages Python, PyTorch, C, C++, Bash, Cuda

Database SQL, MySQL

[Teaching Assistantship](#)

Spring 2023 **AST 4211: Essentials of Astrophysics**, University of Florida.

Instructor: Paul Torrey

- Fall 2020 **AST 2037: Life in the Universe**, University of Florida.
Instructor: Paul Sell
- Fall 2020 **AST 2003: Introduction to the Solar System**, University of Florida.
Instructor: Sarah Ballard
- Fall 2020 **AST 4903: Computational Astrophysics**, University of Florida.
Instructor: Desika Narayanan
- 2017 - 2019 **Math and Physics Tutoring**, University of Oregon.